EDGE COMPUTED WEATHER AND AIR MONITOR

B.TECH. INFORMATION TECHNOLOGY

DESIGN THINKING

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ABSTRACT

Standard IoT weather stations act as passive terminals, relying on cloud latency and generalized web APIs for macro-climate data. This project presents an isolated, Edge-Computed telemetry node powered by an ESP32 microcontroller. By synthesizing real-time data from a BME280, an analog MQ-135, and a physical rain sensor, the system calculates Barometric Tendency and an Empirical Air Quality Score (AQS) entirely offline. Utilizing an asynchronous, non-blocking architecture, the system provides localized "Nowcasting" and event-driven emergency Telegram alerts without cloud-computation delays.

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INTRODUCTION

Applying the core principles of the Design & Thinking framework, we identified a critical vulnerability in modern environmental monitoring. Commercial weather applications pull data from macro-level meteorological APIs, which inherently ignore immediate micro-climates. A city-wide forecast cannot detect a localized volatile gas hazard or a micro-storm forming directly above a specific university laboratory.

The Defined Problem: Cloud-dependent architectures suffer from latency, internet-dependency, and a lack of micro-environmental awareness.

The Proposed Solution: This project was ideated and prototyped to act as a self-sufficient "Edge Node." The objective was to engineer a localized system that processes all mathematical logic on the hardware itself, ensuring zero-latency hazard detection and high-reliability reporting even during network bandwidth fluctuations.

METHODS AND MATERIALS

Hardware Architecture & Methods:

- **The Core Processor:** ESP32 Dual-Core Microcontroller featuring Multi-AP network roaming.
- **The I2C Communications Bus:** To reduce wiring complexity and preserve GPIO pins for interrupts, the BME280 environmental sensor and the OLED Display share a single Inter-Integrated Circuit (I2C) bus on GPIO 21 (SDA) and GPIO 22 (SCL).
- **Analog Sensor Integration:** An MQ-135 Chemi resistor is mapped to the ESP32's 12-bit ADC (Analog-to-Digital Converter) on GPIO 35. A physical droplet sensor utilizes GPIO 34 for immediate precipitation detection.

Software Engineering & Core Logic:

- The system utilizes an asynchronous time-keeping architecture. It synchronizes with the <pool.ntp.org> Network Time Protocol, applying a strict 19,800-second GMT offset to maintain flawless Indian Standard Time (IST) logging.
- **Edge-Computed Barometric Tendency:** The system calculates the immediate rate of change in atmospheric pressure:

$$\Delta P = P_{current} - P_{previous}$$

Sampled over 60-second intervals, a steep negative change in pressure, computationally triggers a "Storm Alert" state, indicating an immediate localized low-pressure vacuum. This pressure delta is mathematically fused with live humidity data to generate a dynamic Rain probability percentage.

- **Empirical Baseline AQS Scoring:** Analog gas sensors cannot accurately output absolute Parts-Per-Million (PPM) without laboratory vacuum calibration. To maintain academic and scientific integrity, this project utilizes an Empirical Baseline. The 12-bit ADC resolution (spanning 0 to 4095) maps ambient air as <1500, while volatile organic compound (VOC) saturation securely triggers hazard states at values >2500.

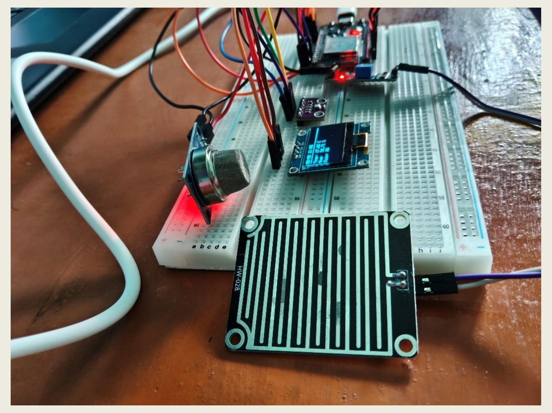


Figure 1. Side view of project

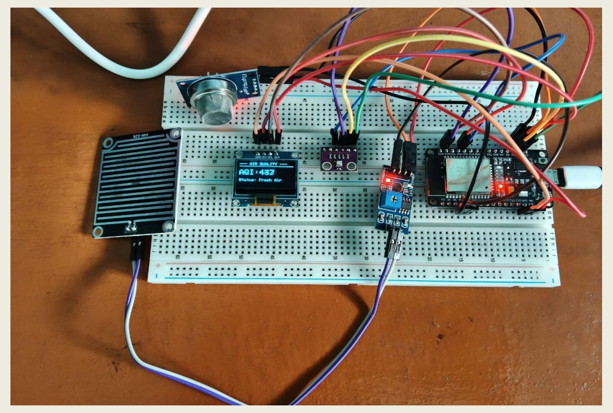


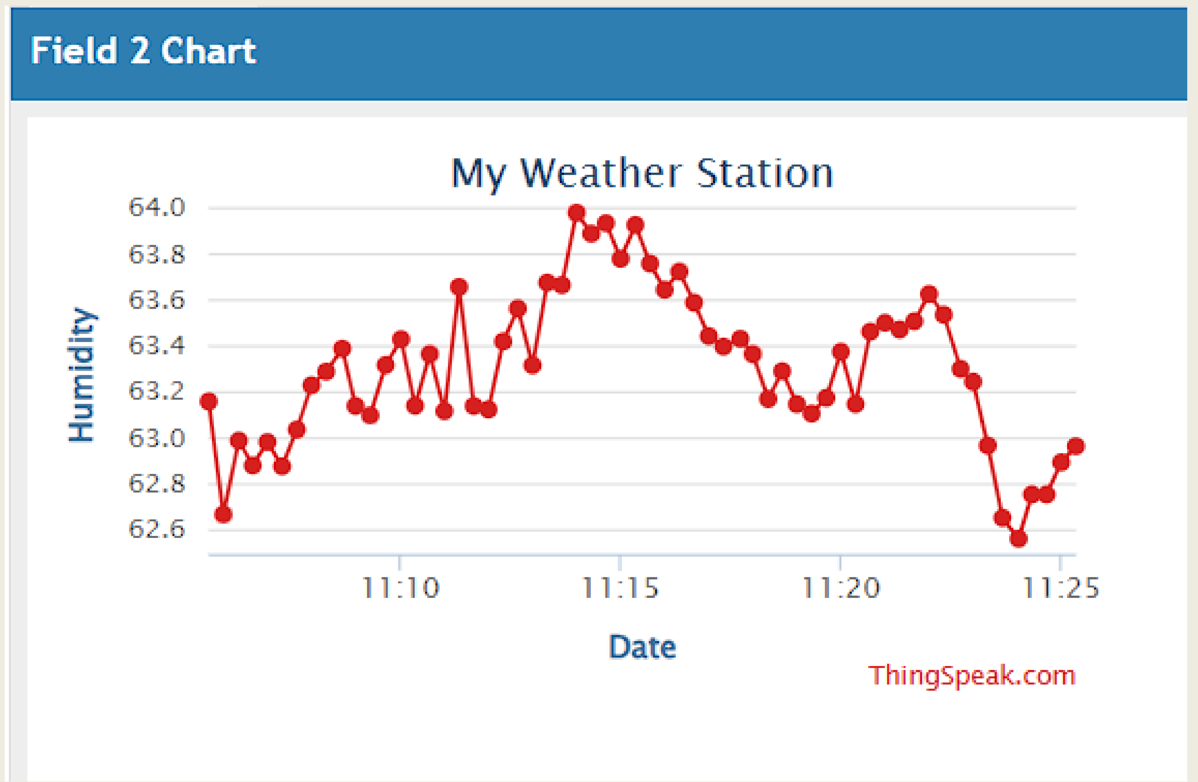
Figure 2. Top view of project

RESULTS

The localized testing phase demonstrated flawless execution of the non-blocking architecture. The OLED UI successfully updates its 3-page rotary display every 3 seconds without interrupting background telemetry.

- **Passive Telemetry:** The system securely POSTs multi-variable time-series payloads to the ThingSpeak REST API every 20 seconds, successfully logging data lakes for future analysis.
- **Active Event-Driven Alerting:** The Telegram Bot integration proved the superiority of edge-interrupts. The system successfully pushes a routine "heartbeat" status report every 2 minutes. However, upon physical moisture detection, the hardware interrupt bypasses the timer to push an immediate **rain alert** to the user's mobile device with near-zero latency.

GRAPH 1: Humidity Graph



CONCLUSIONS

This project successfully bridges the gap between raw data collection and intelligent environmental Nowcasting. By combining non-blocking software architecture, localized empirical mathematics, and sensor fusion, the prototype successfully functions as a robust, low-cost Edge-IoT device capable of autonomous environmental analysis and instantaneous emergency communication.

FUTURE SCORE

- **TinyML (Machine Learning on the Edge):** Transitioning from static deterministic logic to pattern recognition. By harvesting the accumulated ThingSpeak CSV data, a quantized TensorFlow Lite neural network can be trained to recognize seasonal micro-climate patterns and flashed directly to the ESP32.
- **Analog Wind Vane Integration:** Engineering a custom wind-direction sensor utilizing a low-cost 10k rotary potentiometer. By mapping the 0-4095 voltage drop to a 360-degree software compass array, the system could seamlessly unlock the parameters required to run the fully modified Zambretti Algorithm entirely on the edge.

REFERENCES

- https://iopscience.iop.org/article/10.1088/1757-899X/537/3/032086/pdf